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Introduction

The objective of this experiment was to study the behavior of common emitter transistor AC amplifiers. The experiment also focused on the effect of the emitter capacitor on voltage gain and Q – point.

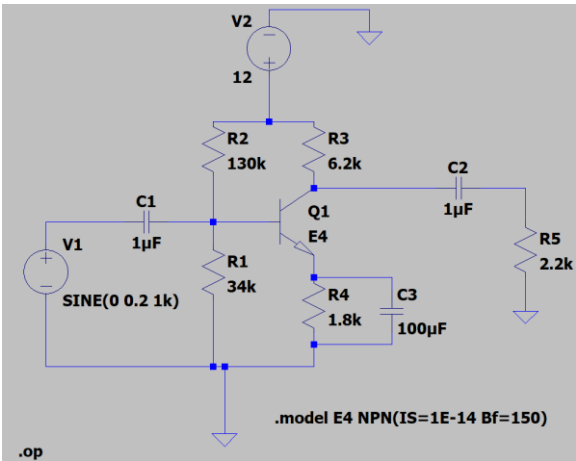
Experimental Procedure

1. Initial Biasing: A CE amplifier was designed using the parameters of $V_{cc} = 12\text{ V}$, $R_c = 6.2\text{ k ohms}$, $R_e = 1.8\text{ k ohms}$, $R_L = 2.2\text{ k ohms}$, and with R_1 and R_2 chosen based on calculations ($R_1 = 164\text{ k ohms}$, $R_2 = 33\text{ k ohms}$) to acquire $I_{cq} = 1\text{ mA}$.
2. Max Unclipped Output (with C_e): A 5 kHz sinusoidal signal is applied, and the input amplitude is increased until the output waveform began to clip. The maximum unclipped output was then measured.
3. Effect of C_e Removal: The capacitor C_e was removed, and the maximum unclipped output is observed.
4. Optimized Biasing: R_1 and R_2 were replaced to center the Q-point on the AC load line, and the maximum unclipped output was measured.
5. Optimized Biasing without C_e : Step 4 was then repeated with capacitor C_e removed.
6. Impact of R_c Variation: Resistor R_c was changed to 3.2 k ohms, and then I_{cq} was measured.

Computer Simulation

Initial Bias Configuration (Parts 1 & 2)

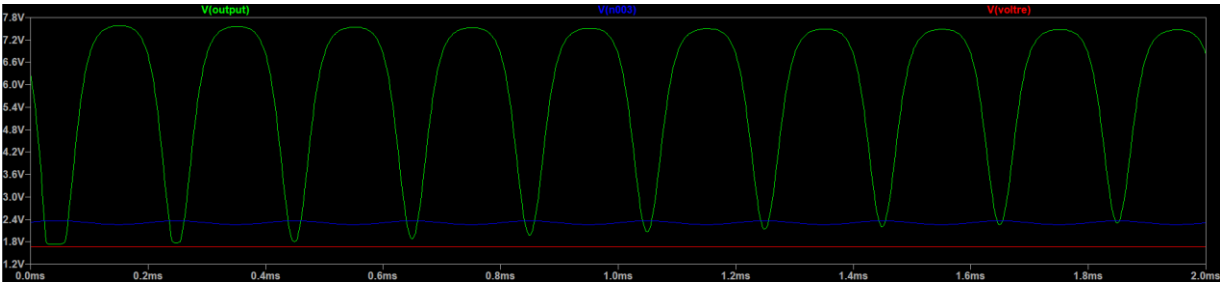
The initial circuit schematic and the Dc operating point data is shown below. The results confirmed that I_{cq} is approximately 1 mA.



--- Operating Point ---		
V(n001) :	12	voltage
V(n002) :	6.28832	voltage
V(n003) :	1.38343e-14	voltage
V(n004) :	0	voltage
V(n005) :	2.32228	voltage
V(n006) :	1.66928	voltage
I(C1) :	-2.32228e-18	device_current
I(C2) :	-6.28832e-18	device_current
I(C3) :	1.66928e-16	device_current
I(R1) :	6.83024e-05	device_current
I(R2) :	7.4444e-05	device_current
I(R3) :	0.000921239	device_current
I(R4) :	0.00092738	device_current
I(R5) :	6.28832e-18	device_current
I(V1) :	2.32228e-18	device_current
I(V2) :	-0.000995683	device_current
Ib(Q1) :	6.14159e-06	device_current
Ic(Q1) :	0.000921239	device_current
Ie(Q1) :	-0.00092738	device_current
Is(Q1) :	0	device_current

Simulation Schematic

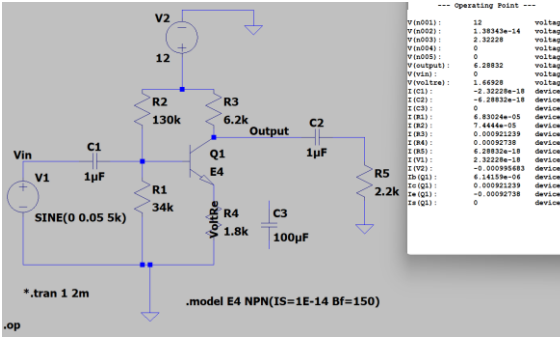
Operating Point Data



Simulation waveform

Optimized Bias Configuration

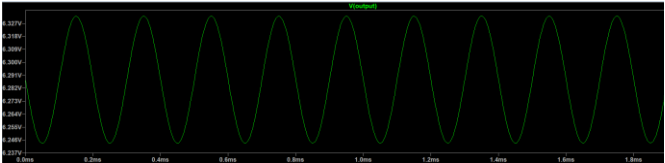
The simulation below shows the increased output and the effects of varying R_c and C_e .



Schematic

--- Operating Point ---

V(in001)	12	voltage
V(in002)	1.38343e-14	voltage
V(in003)	2.32228	voltage
V(in004)	0	voltage
V(in005)	0	voltage
V(out001)	6.28832	voltage
V(vin)	0	voltage
V(vout)	1.46928	voltage
I(C1)	-2.32228e-18	device
I(C3)	-6.28832e-18	device
I(C3)	0	device
I(R1)	6.20204e-05	device
I(R2)	7.4444e-05	device
I(R3)	0.00021229	device
I(R4)	0.0002708	device
I(R5)	6.28832e-18	device
I(V1)	2.32228e-18	device
I(V2)	-5.00095683	device
IB(Q1)	6.16158e-06	device
IC(Q1)	0.00021229	device
Ie(Q1)	-5.0006739	device
Ia(Q1)	0	device

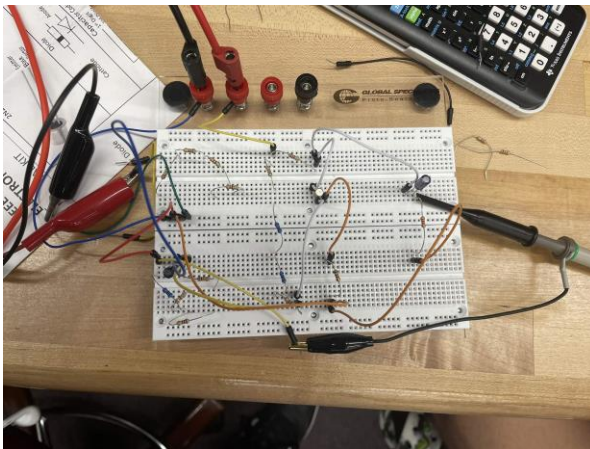


Waveform

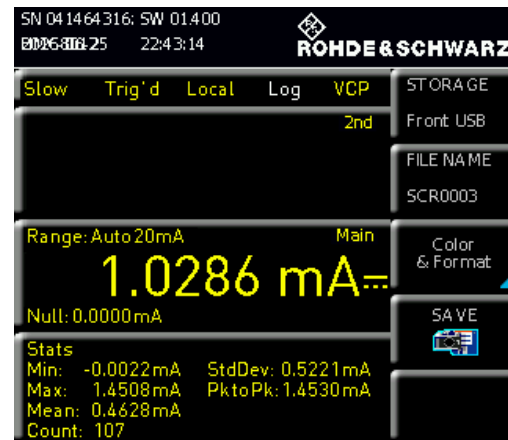
Results and Discussion

Initial Circuit Performance

The initial circuit measured quiescent current I_{cq} was 1.0298 mA, which is the target design of 1 mA and the simulation results.

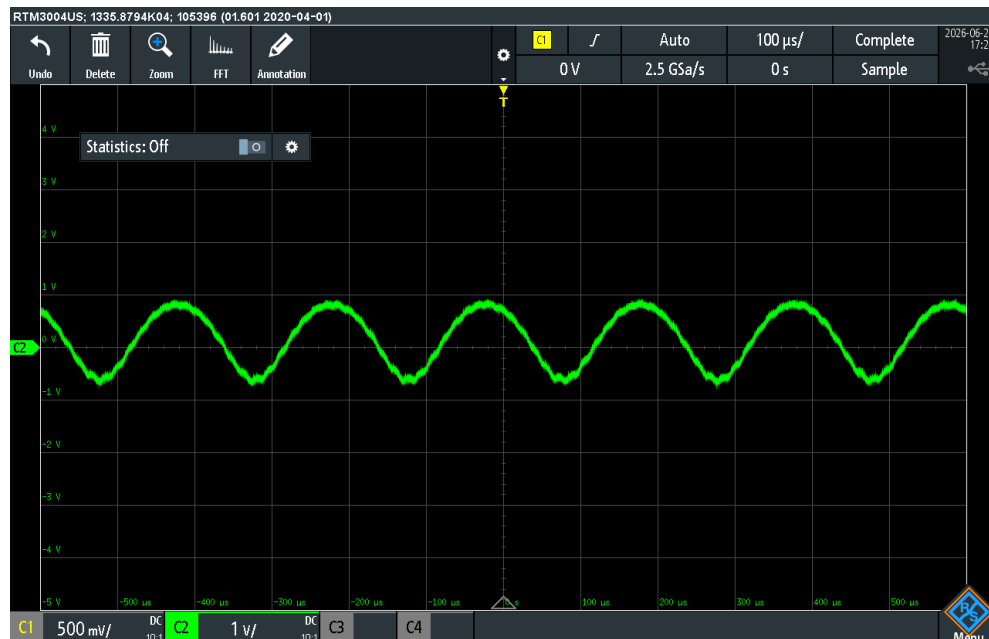


Breadboard assembly

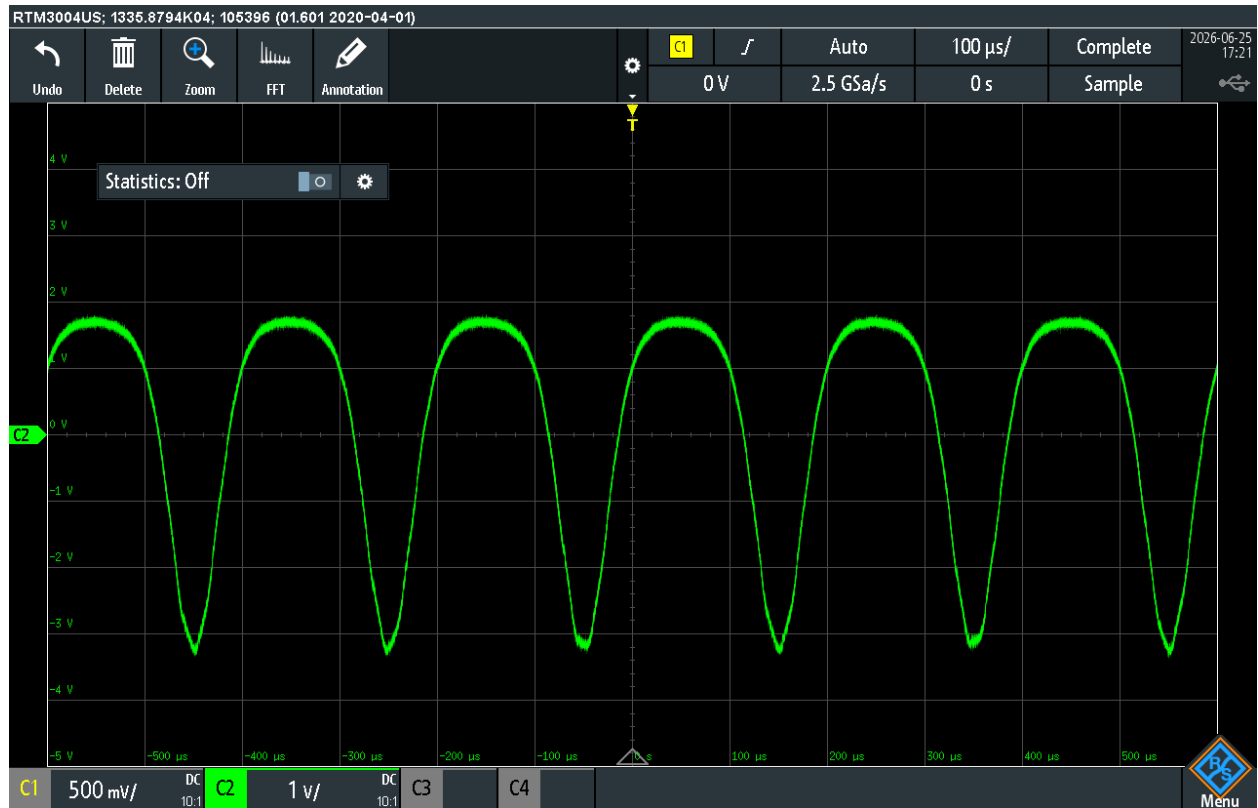


I_{cq} approx. 1 mA

With the capacitor C_e included, the amplifier showed high gain. The maximum unclipped output waveform is shown.



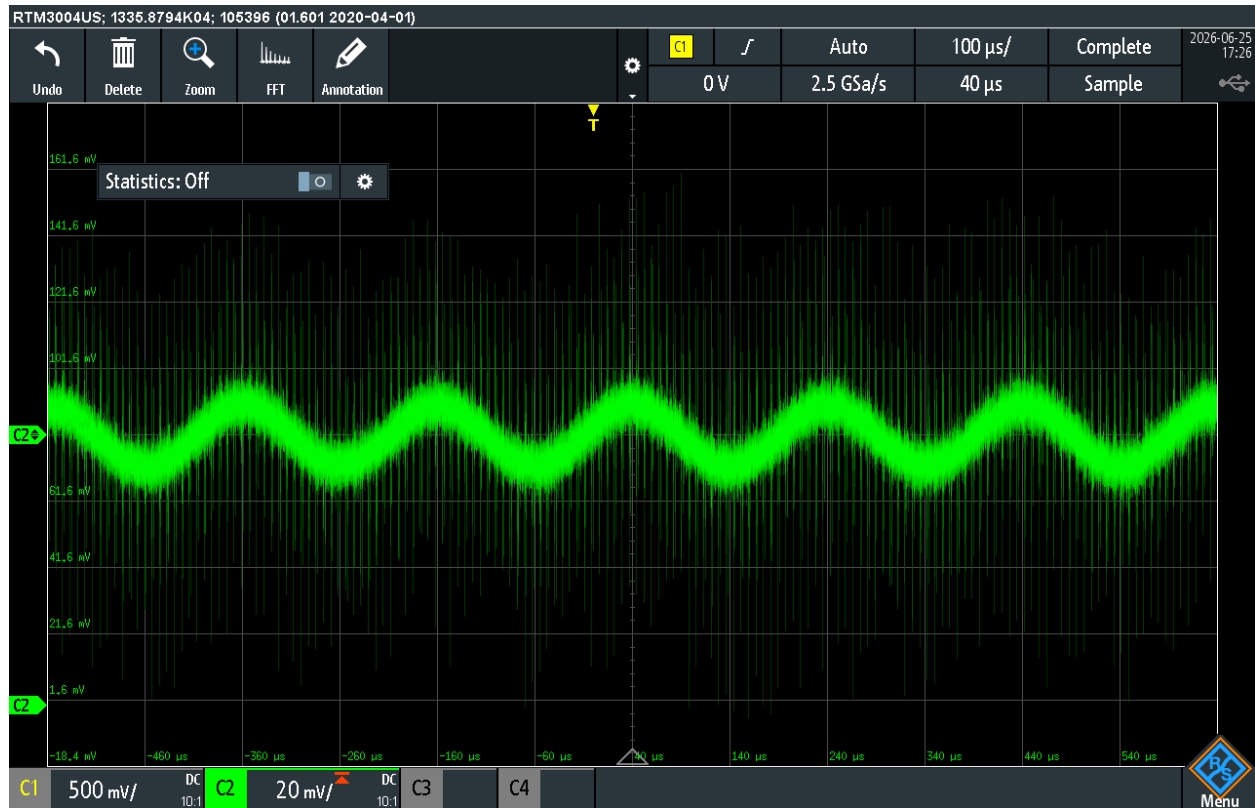
Unclipped



Clipped

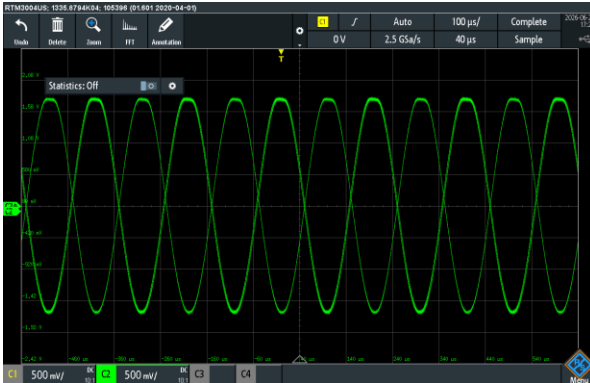
Effect of C_e Removal

By removing C_e , the gain drops, and the maximum unclipped output was reduced.

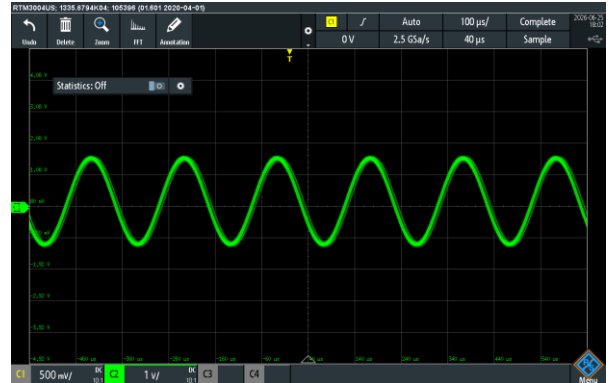


Optimized Q-point Performance

To maximize the output, the Q-point was shifted by making R1 to 100 k ohms and R2 to 34 k ohms. Shown below is the resulting waveforms. The output is visibly larger in the previous configuration. Removing C_e reduced the gain of our circuit as expected.



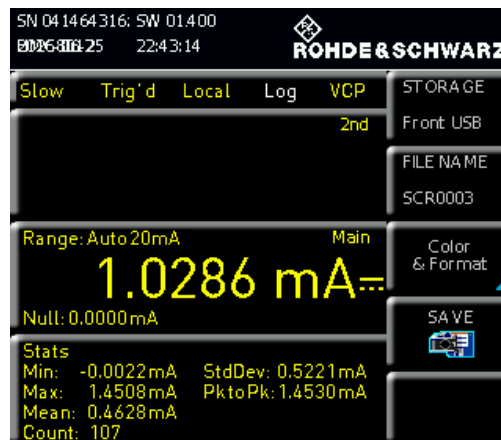
Bias with C_e



Bias without C_e

Bias Stability and R_c Variation

Changing the R_c did not have a significant impact on the collector current. The measured current was 1.0286 mA. This confirms that the voltage divider bias circuit provides exceptional DC stability.



PreLab

1)

$$R_{BB} = R_1 // R_2 \quad R_{BB} = 0.1(151)(1.8k)$$

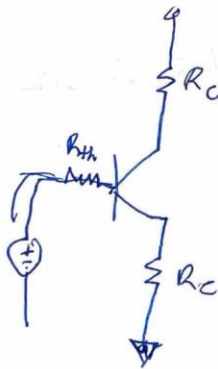
$$R_{BB} = 27,180$$

$$I_c = \frac{V_{cc} - V_c}{R_c} \rightarrow 1mA = \frac{12 - V_c}{6.2k} \Rightarrow V_c = 5.8V$$

$$I_c = \beta I_B \Rightarrow 1mA = 150 I_B$$

$$I_B = 6.67 \mu A$$

$$V_{th} = \frac{V_{cc}}{R_1} \cdot R_{th} = \frac{12}{R_1} \cdot 27,180$$



$$V_{th} = \frac{1mA}{150} (27,180) + \left(\frac{151}{150} \right) (1mA)(1.8k)$$

$$V_{th} = 1.9932V$$

$$1.9932 = \frac{12}{R_1} \cdot 27,180$$

$$27,180 = \frac{R_2 \cdot 463,636}{R_2 + 163,636} \rightarrow R_2 \approx 32,594 \Omega$$

$$R_1 \approx 163,636 \Omega$$

2)

$$V_{CEQ} = V_{CC} - I_C R_C - I_E R_E$$

$$= V_{CC} - I_C R_C - \left(\frac{\beta+1}{\beta}\right) I_C R_E$$

$$V_{CE} = 12 - (1 \text{ mA})(6.2 \text{ k}) - \left(\frac{151}{150}\right)(1 \text{ mA})(1.8 \text{ k})$$

$$V_{CE} \approx 4 \text{ V} \quad \rightarrow Q \approx (1 \text{ mA}, 4 \text{ V})$$

$$V_E = I_E R_E = \left(\frac{\beta+1}{\beta}\right) I_C R_E = \left(\frac{151}{150}\right)(1 \text{ mA})(1.8 \text{ k})$$

$$V_E = 1.812 \text{ V}$$

$$R_{AC} = R_L \parallel R_C = 6.2 \text{ k} \parallel 2.2 \text{ k} = 1.6238 \text{ k} \Omega$$

$$V_{out} = 1 \text{ mA} \cdot 1.624 = 1.624 \text{ V}$$

$$= 1.624 \cdot 2 = 3.248$$

$$5) V_{th} = V_E + 0.7 = 1.832 + 0.7 = 2.512$$

$$2.512 = 12 \cdot \frac{R_2}{R_1 + R_2} \Rightarrow 2.512 = \frac{12}{R_1} \cdot R_{th}$$

$$\Rightarrow \frac{12}{R_1} \cdot 27180 \Rightarrow R_1 \approx 100K$$

$$2.512 = 12 \cdot \frac{R_2}{100K + R_2} \Rightarrow R_2 \approx 26,475K$$

$$6) I_c \approx \frac{V_{BB} - V_{BE}}{\frac{R_{th}}{\beta} + R_E} = \frac{2.512 - 0.7}{\left(\frac{27180}{150}\right) + 1.8K} \approx 1mA$$

$$V_C = 12 - (1mA)(3.2K) = 8.8V$$

$$V_{CE} = V_{CC} - I_c R_C - \left(\frac{\beta + 1}{\beta}\right) I_c R_E$$

$$V_{CE} = 12 - (1mA)(3.2K) - \left(\frac{151}{150}\right)(1mA)(1.8K)$$

$$V_{CE} = 6.988V$$

Conclusion

This experiment validated the concept of common emitter AC amplifiers. It shows that the capacitor C_e is necessary for achieving exceptional voltage gain, while removing the component improves stability. The stability of the voltage divider bias was also confirmed by measuring the collector current remaining constant despite changes in resistance R_c .